Reg. No. | | | | | | | | | | | [14AE55]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2017

HEAT & MASS TRANSFER

Time:3 Hours **Maximum Marks:70**

Instructions: i. Answer any **FIVE** full Questions choosing atleast two from each part.
ii. Missing data may be suitably assumed.

PART-A

1. a. Derive the one-dimensional heat conduction equation in cylindrical coordinate system for constant thermal conduction. **10**

b. The roof of a house is 6m long, 8m wide and 0.25m thick and is made of a flat layer of concrete whose thermal conductivity is k=0.8 W/(m·K). One night, the temperatures of the inner and outer surfaces of the roof are measured to be 15°C and 4°C, respectively, for a period of 10 hours. Determine the rate of heat loss through the roof that night and the amount of heat loss during a 10 hour period. **04**

2. A 2-kW resistance heater wire with thermal conductivity $k = 15$ W/(m·K), diameter of $D = 4$ mm and length $L = 0.5$ m, is used to boil water. If the outer surface temperature of the resistance wire is $T_s = 105$°C,

i. Obtain a relation for the temperature distribution. **12**

ii. Determine the temperature at the centreline of the wire. **02**

3. a. In the equation for instantaneous temperature distribution,

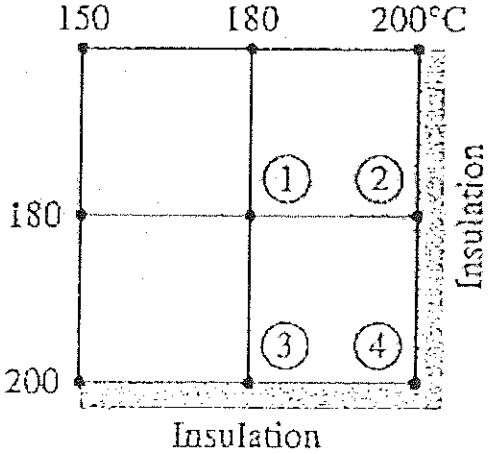
i. Explain the significance of time constant **03**

ii. Derive the dimensionless form of the exponent **03**

b. A stainless steel rod of outer diameter 1 cm originally at a temperature of 320°C is suddenly immersed in a liquid at 120°C for which the convective heat transfer coefficient is 100 W/m^2·K. Determine the time required for the rod to reach a temperature of 200°C. $\rho = 7800$ kg/m^3, $c = 460$ J/(kg·K), $k = 40$ W/(m·K) **08**

4. a. Consider steady two-dimensional heat transfer in a long solid bar of square cross section. The measured temperatures at selected points of the outer surfaces are as shown in Figure 4(b). $k = 20$ W/(m·°C), and there is no heat generation. With a mesh size of $\Delta x = \Delta y = 1$ cm, determine the temperatures at nodes 1, 2, 3 and 4. **12**

b. In the numerical methods, the _____ equations are replaced by algebraic Equations. In the finite difference method, this is done by replacing the _____ by differences. **02**



PART - B

5. a. Air at 27°C and 1 atm flows over a flat plate at a speed of 2 m/s. The plate is 2m long and 1.5 cm wide. If the plate is heated over its entire length to a temperature of 60°C, calculate the heat transferred in the first 40 cm of the plate. **08**

T(°C)	ν (m^2/s)	k (W/(m·°C))	Pr
27	1.59×10^{-5}	0.0263	0.7
43.5	1.736×10^{-5}	0.02749	0.7

b. The steady state temperature distribution in a wall is $T = 300 - 3050x^2$, where x (in meter) is the position in the wall and T is the temperature in (°C). Thickness of the wall is 0.2 m and the thermal conductivity of the wall is 1.2 W/m·°C. If the wall dissipates heat to the ambient at 30°C, calculate the heat transfer coefficient at the surface of the wall. **06**

6. a. State and prove Kirchhoff's law of radiation. **06**

b. Two large plates are at 1000°K and 800°K. Determine the heat exchange per unit area when, **05**

i. The surfaces are black.

ii. The hot surface has an emissivity of 0.9 and cold 0.6.

c. Explain the concept of a perfect black body. **03**

7. a. A counter flow heat exchanger of heat transfer area A=12.5 m^2 is to cool oil (c_{ph} = 2000 J/(kg·°C)) with water (c_{pc} = 4170 J/(kg·°C)). The oil enters at $T_{h,i}$ = 100°C and $\dot{m}_h$ = 2 kg/s, while water enters at $T_{c,i}$ = 20°C and $\dot{m}_c$ = 0.48 kg/s. The overall heat transfer coefficient is U_m = 400 W/(m^2·°C). Calculate the exit temperature of water and the total heat transfer rate. **08**

b. Determine the overall heat transfer coefficient U_o based on the outer surface of a 2.54 cm O.D. and 2.286 cm I.D. heat exchanger tube ($k = 102$ W/m·K), if the heat transfer coefficients at the inside and outside of the tube are $h_i = 5500$ W/m^2·K and $h_o = 3800$ W/m^2·K, respectively and the fouling factors are $R_{f,o} = R_{f,i} = 0.0002$ m^2/W·K. **06**

8. a. Explain the significance of **06**

i. Schmidt number

ii. Levis number

b. A vessel contains a binary mixture of oxygen and nitrogen at 27°C with partial pressures in the ratio 0.21 and 0.79 respectively. The total pressure of the mixture is 1.1bar. Calculate the mass fraction of oxygen. **08**